

COMPOUND MICROSCOPE

A PROJECT ANALYSIS

SREE VED SOMA:-MS24BTECH11031

DEVANANDA ASHOKAN:- MS24BTECH11008

GAYATRI REKNER:- MS24BTECH11028

LAHARI YADA:- MS24BTECH11033

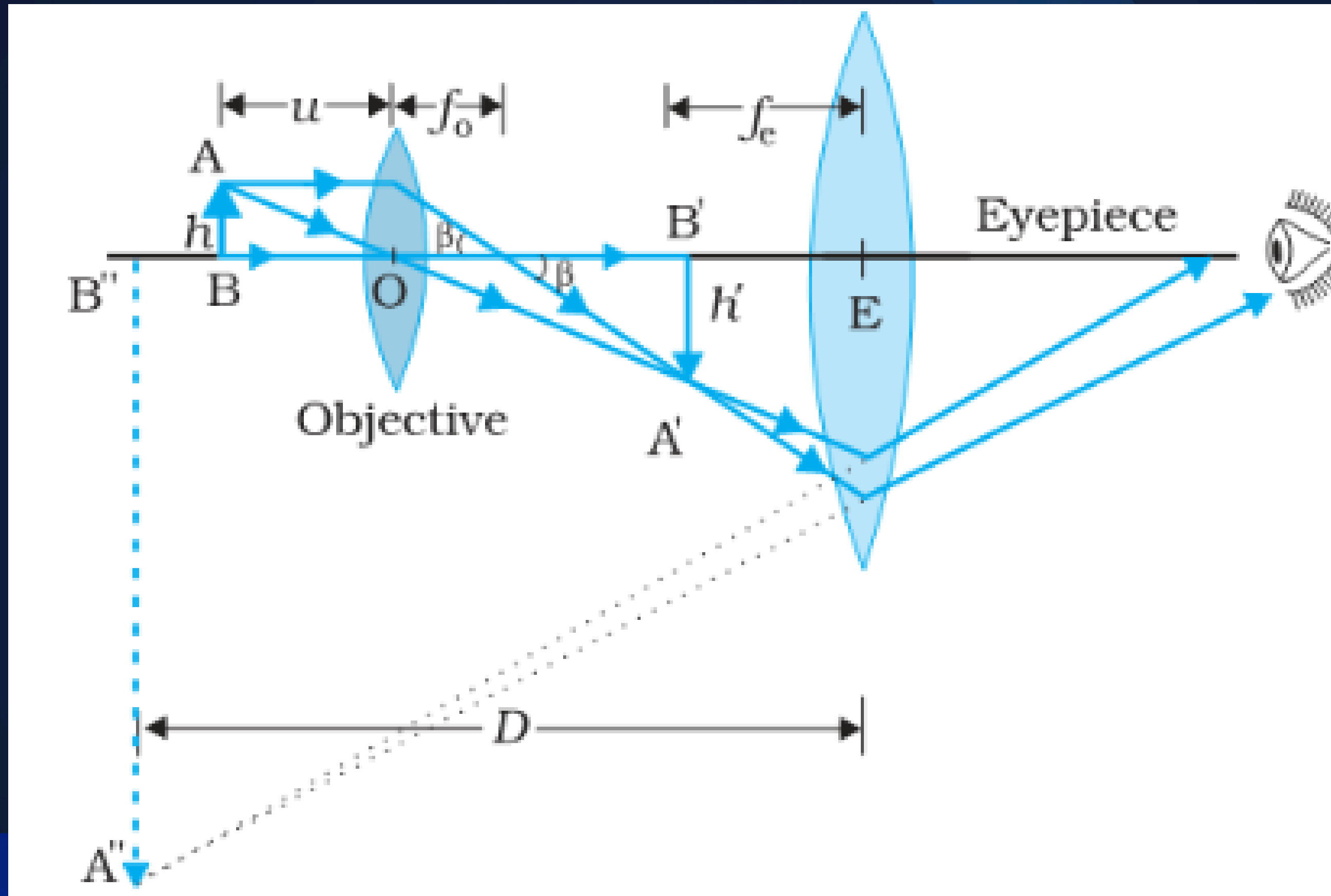
JOSEPH DANIEL:- MS24BTECH11020

AN OVERVIEW

- *A compound microscope uses two or more convex lenses to achieve high magnification and sharp resolution.*
- *The objective lens, close to the specimen, forms a real, inverted, and magnified intermediate image.*
- *Typical total magnification ranges from 40× to 1000×, ideal for studying microscopic cells and structures.*
- *The eyepiece lens, nearer to the observer's eye, further magnifies this image to create a virtual, large final image.*



IMAGE FORMATION



WORKING

Object Placement (Specimen)

Just beyond the
focal point of the
objective lens

Intermediate / Primary Image (Formed by Objective)

On the other side of
the objective lens,
at a distance inside
the microscope tube
but closer to the
eyepiece side.

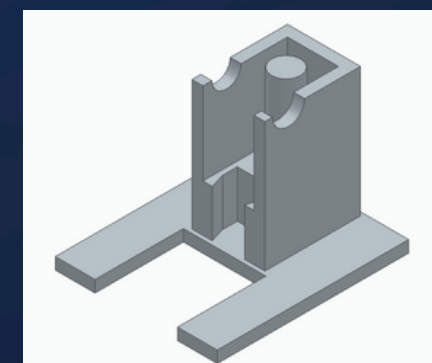
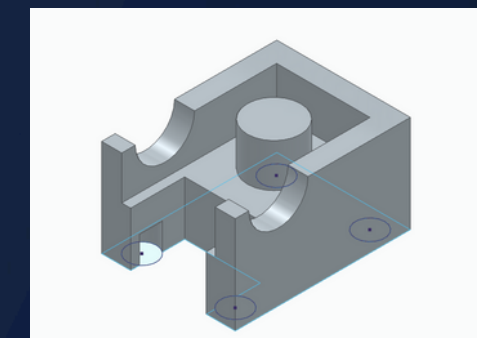
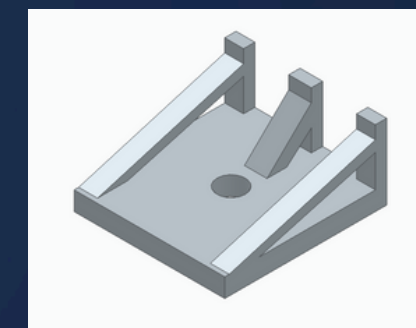
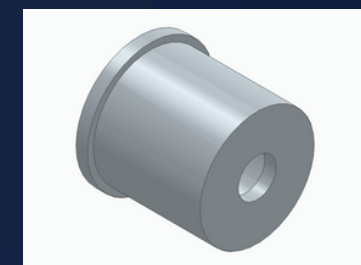
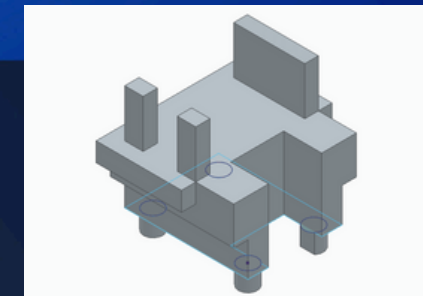
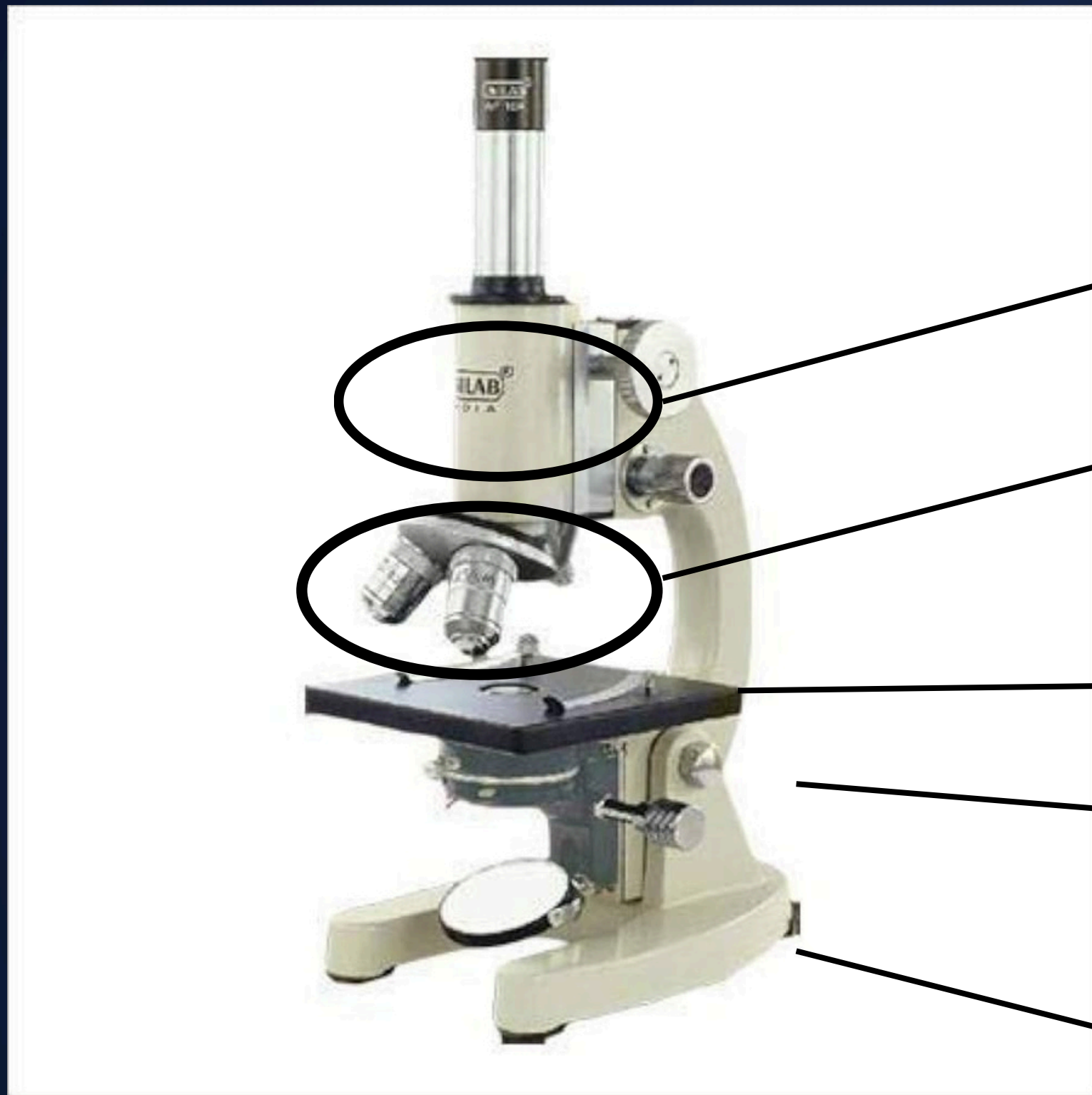
Eyepiece Lens Position

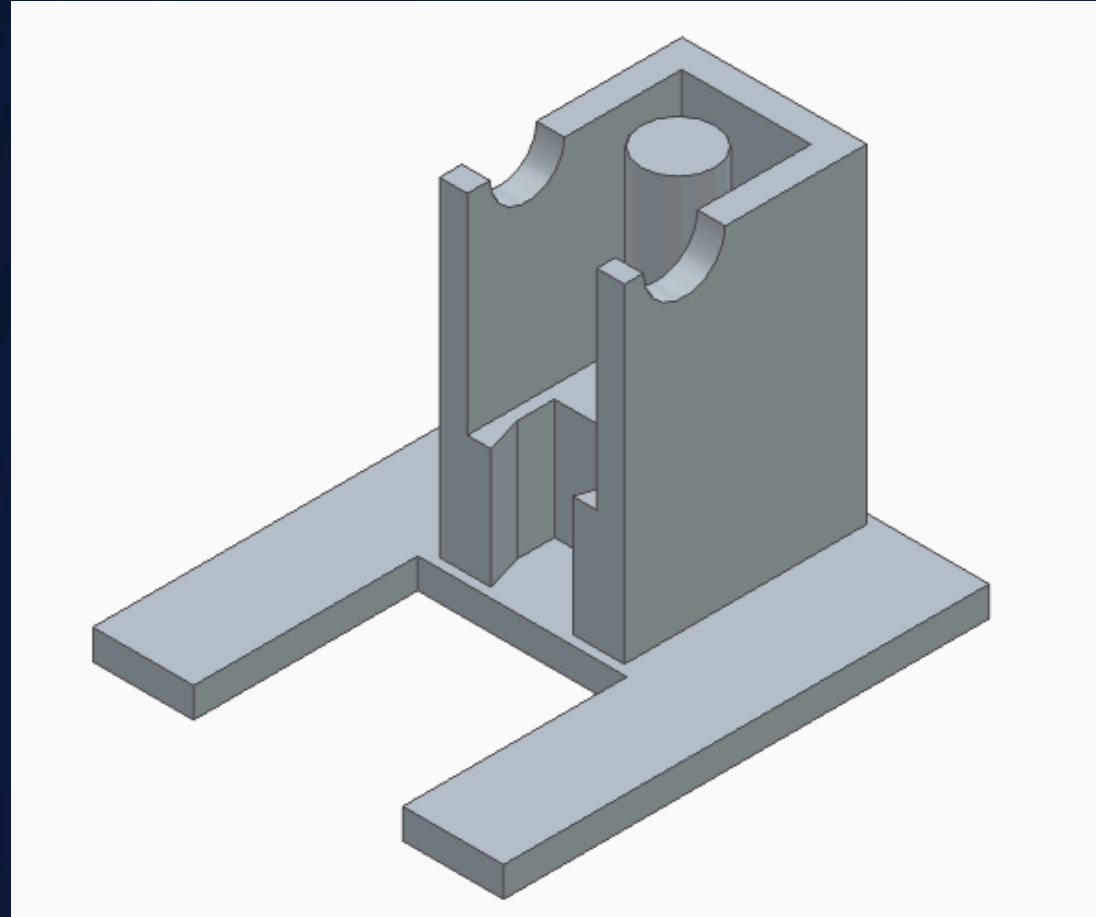
The eyepiece lens is
placed so that the
intermediate image
lies within its own
focal length —or
slightly inside,
depending on
viewing comfort.

Final Image (Formed by Eyepiece)

The final image is virtual,
inverted, and highly
magnified. Typically, it
appears at "infinity" (for
relaxed viewing) or at the
"least distance of distinct
vision" (about 25 cm from
the eye), which is where your
eyes can view it
comfortably without strain.

OUR PARTS





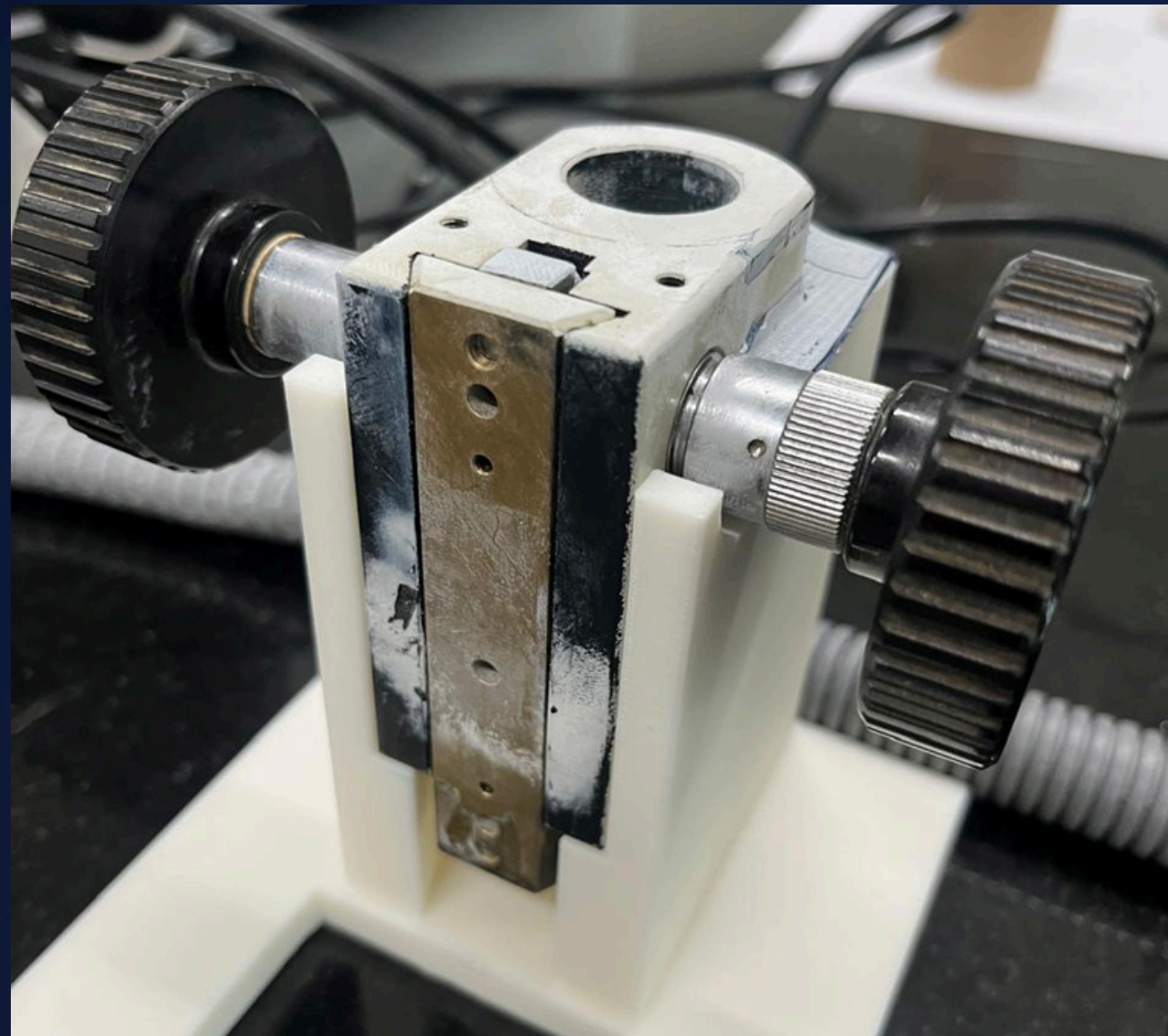
MAIN BASE

modelled after measuring racket & pinion part and base light comfortably, and desgined in a way to integrate them comfortably



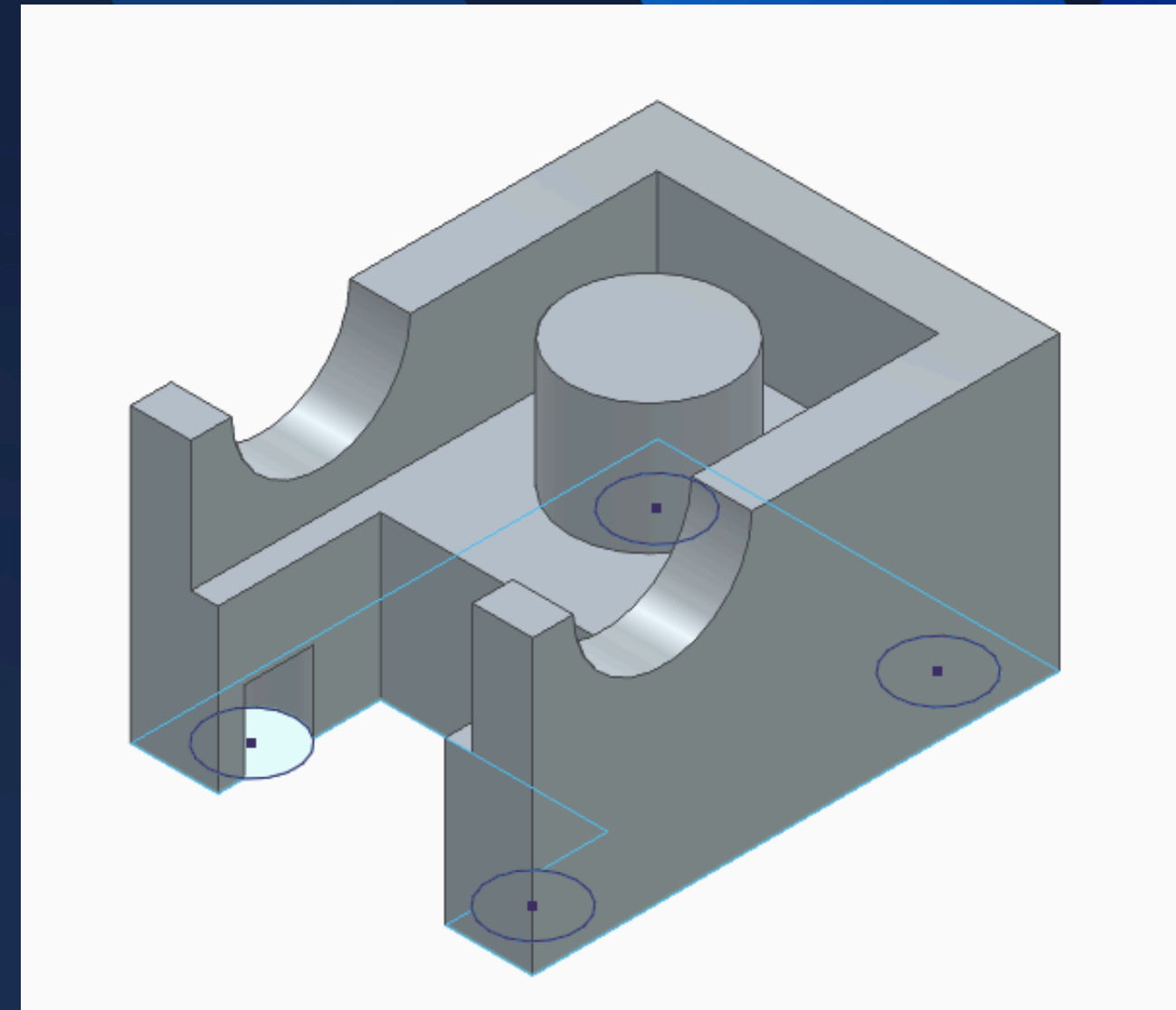
BOTTOM LIGHT

light passes through the specimen from below and enters objective best for thin and transparent samples



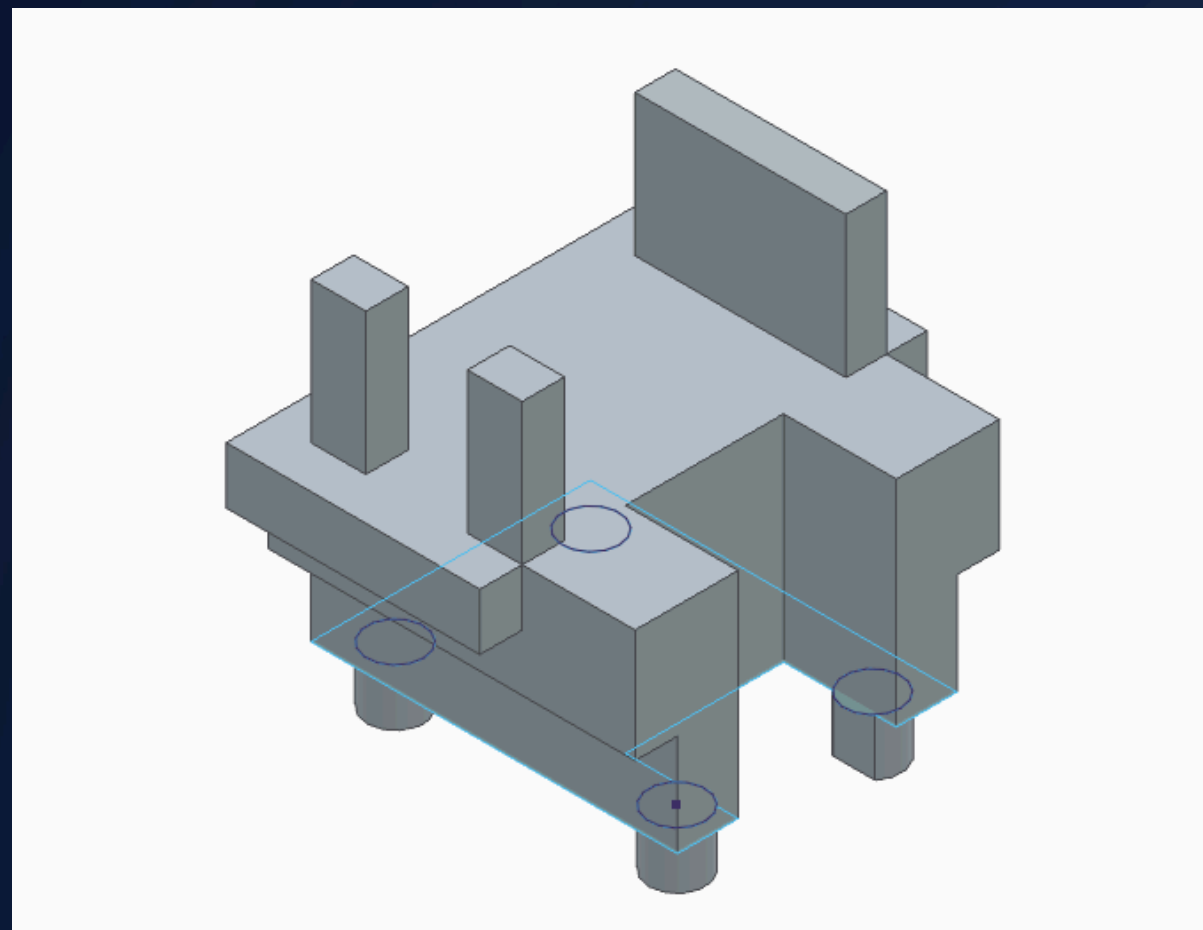
RACKET AND PINION

Salvaged part that converts rotary knob movement into smooth linear stage motion. used for moving the sample table vertically



R & P COVER

Covers the top part of the R&P, perfect fit including special moulds for rotating knobs of the R&P part as well as the top light holder



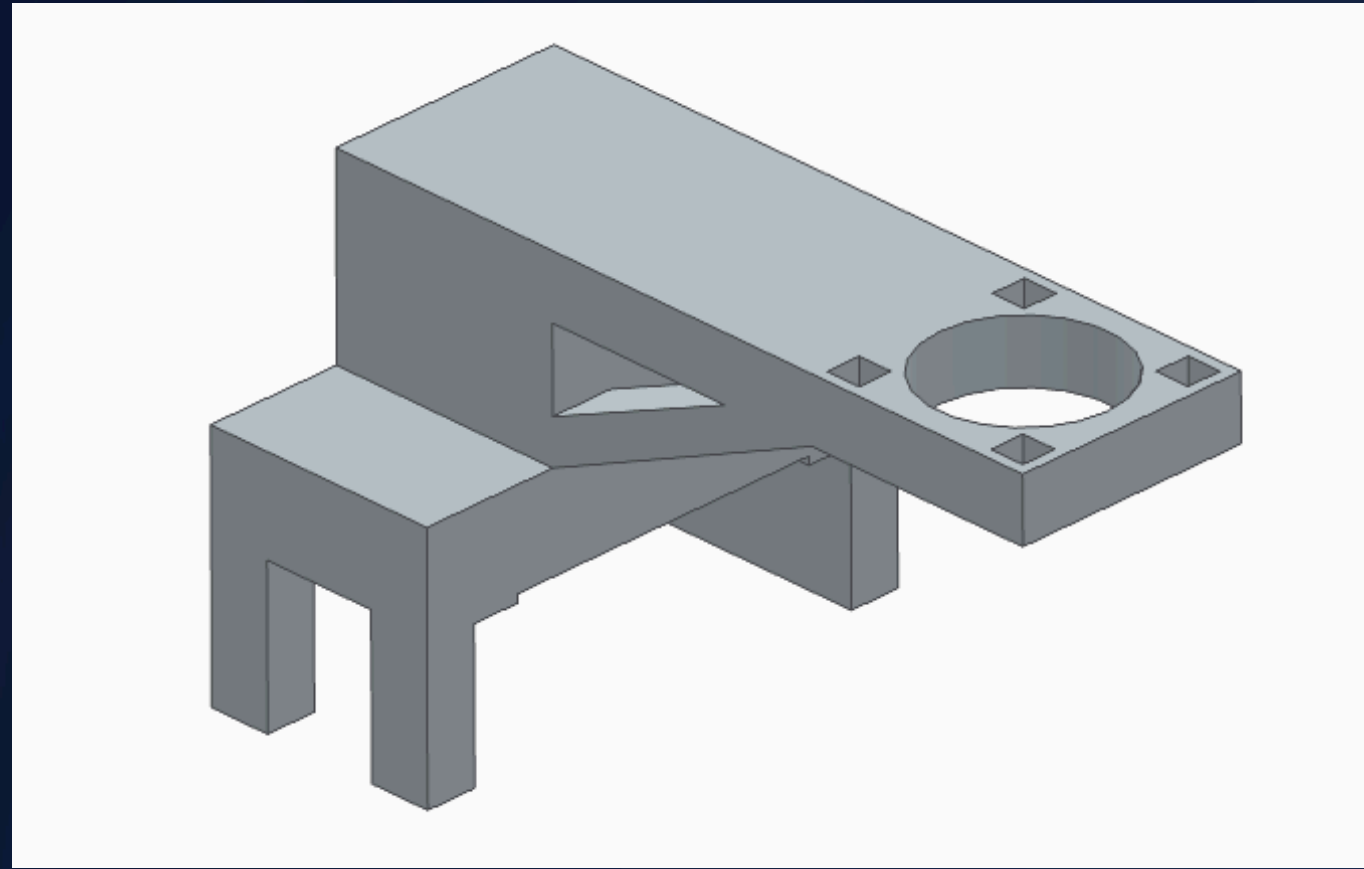
LED RING HOLDER

Holds the led ring light snugly, has been modeled to include space for power port of light, fits perfectly upon the R&P cover with the 4 poles in the holes



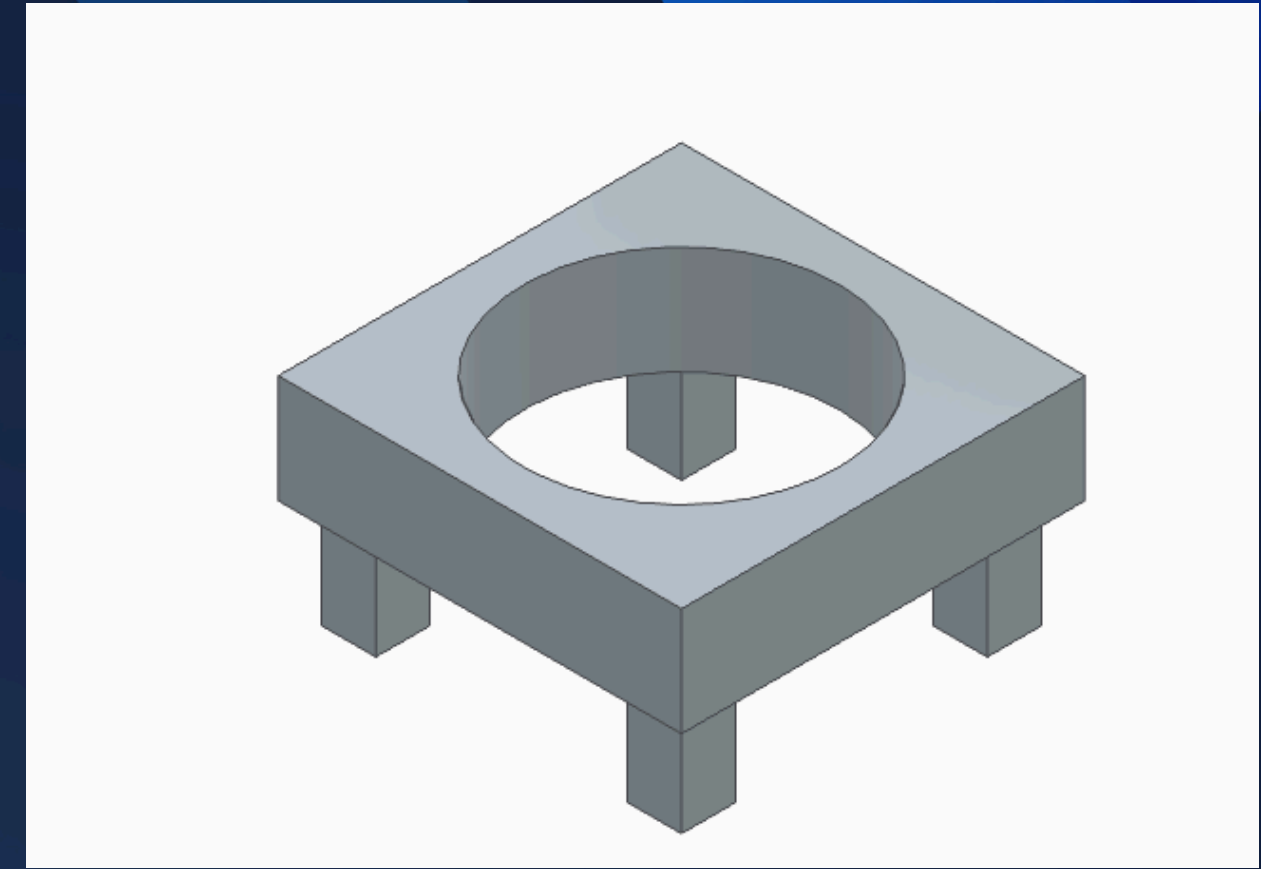
MICROSCOPE LED RING ILLUMINATOR

Another salvaged part which provides illumination from above the sample, causes bright field imaging, 12V with 68 LEDs with intensity varying knob and switch



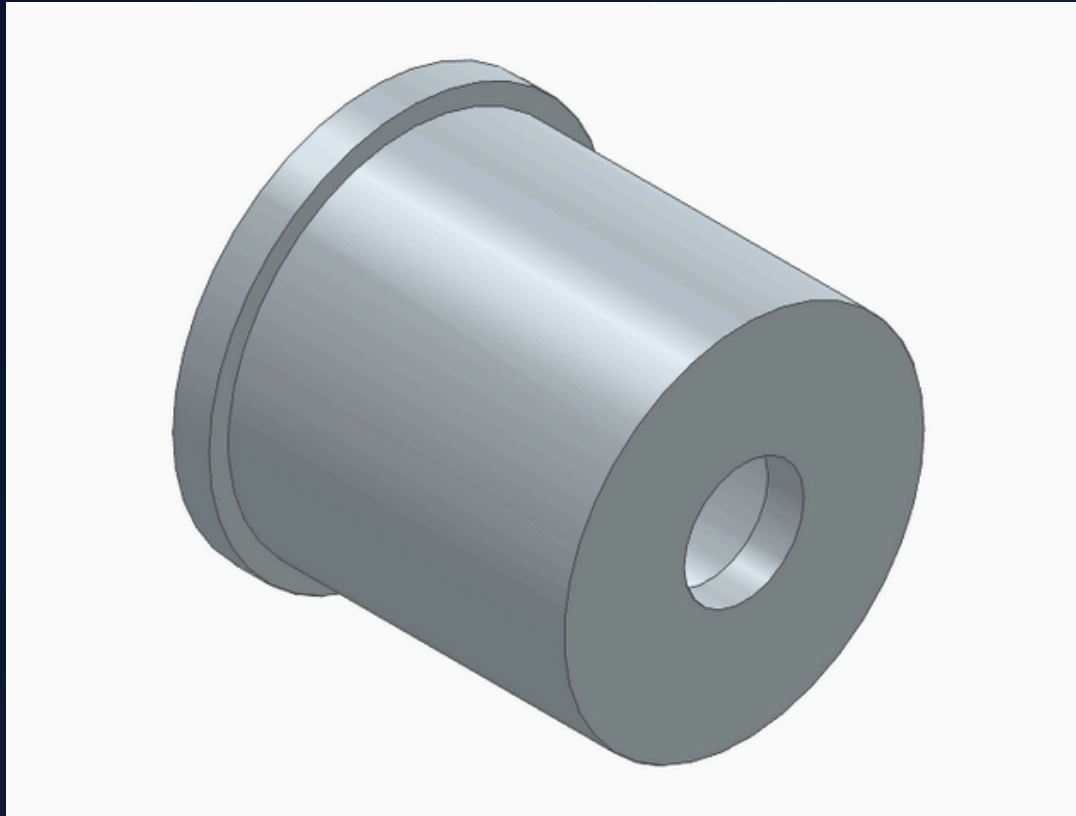
TUBE MOUNTING BRACKET

Attaches on top of the led ring holder and led ring, allows tube to pass through freely

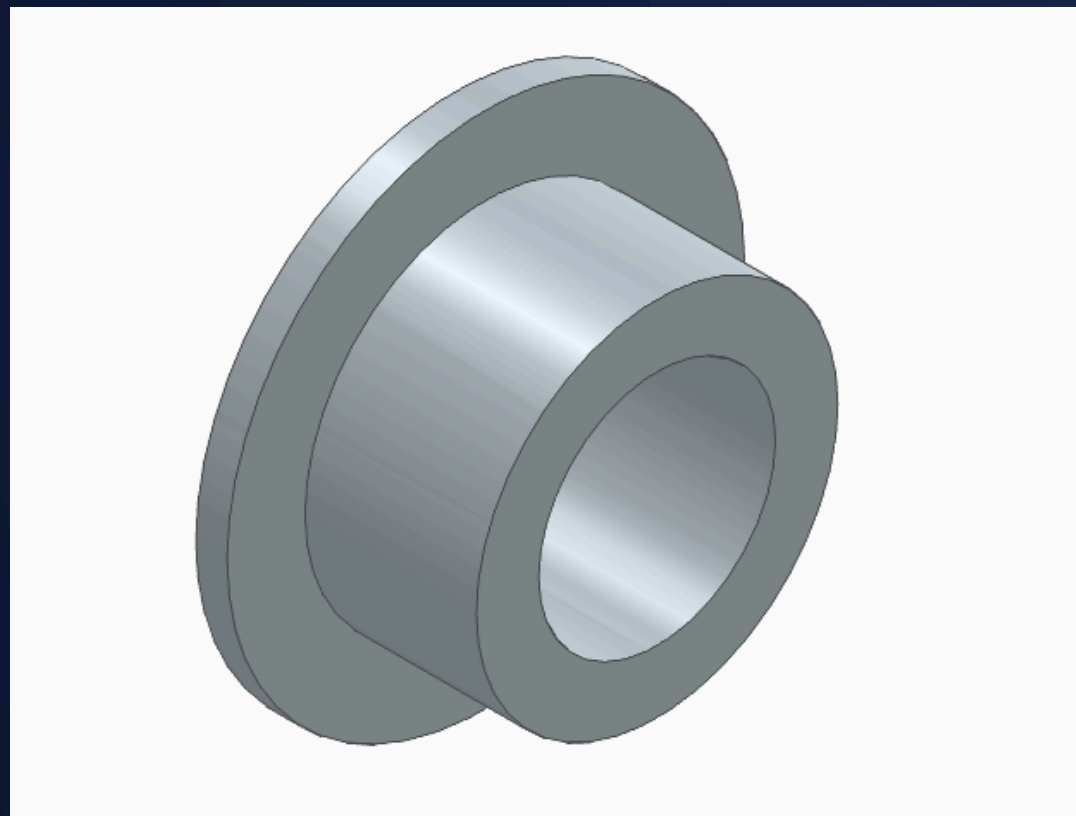


TUBE TIGHT-FIT HOLDER

Attaches on top of the tube mounting bracket, allows tight fit of tube such that exact height of body tube can be fixed



OBJECTIVE AND IT'S PARTS



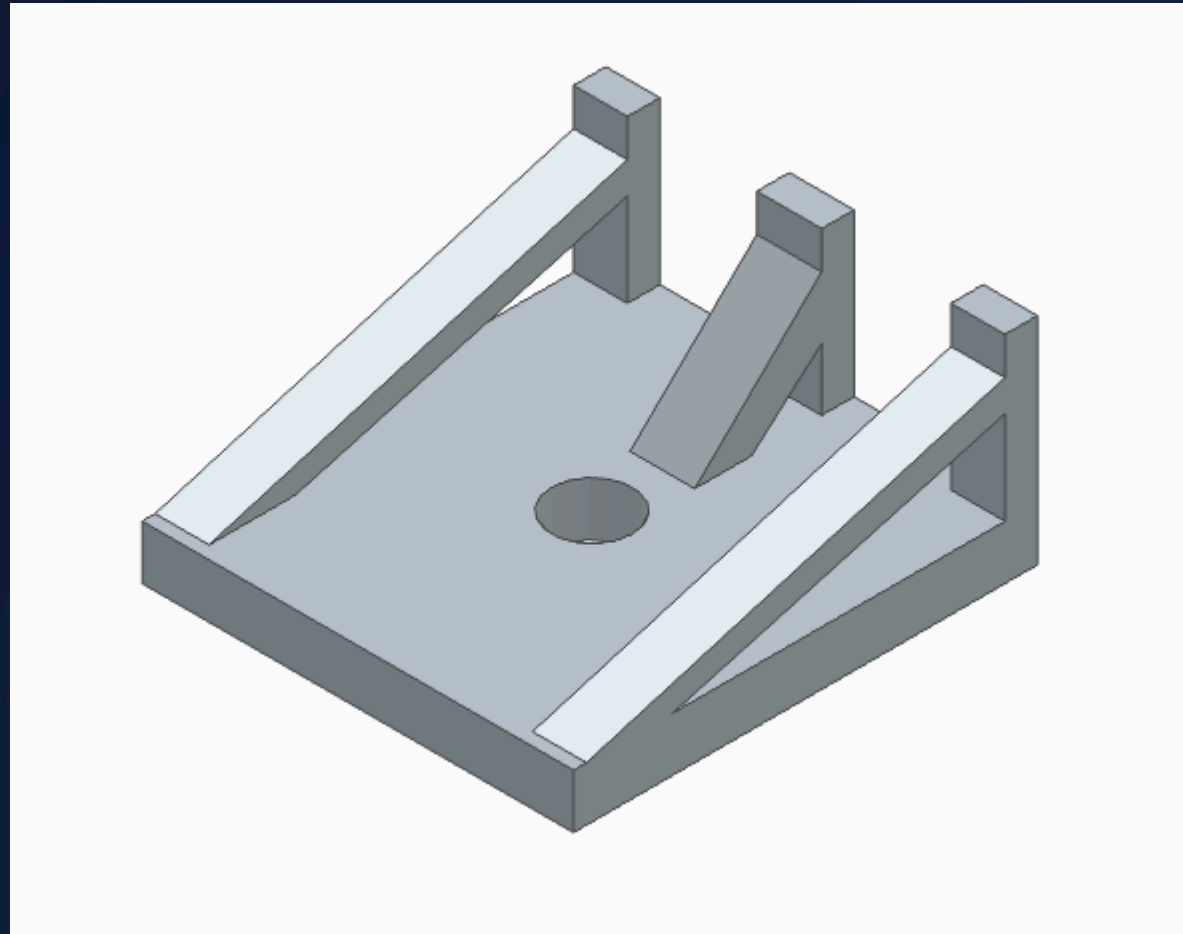
The objective is a convex lens which is enclosed carefully inside both the parts which are then stuck and the top part is then slid into one end of the body tube.

Magnification - 4×

Numerical Aperture - 0.10

Tube Length - 160 mm

Cover Glass Thickness - 0.17 mm



SAMPLE TABLE

sample is placed on the table, hole in middle allows light from bottom source as well, contains ribs to enhance strength and rigidity, help prevent deformation over time

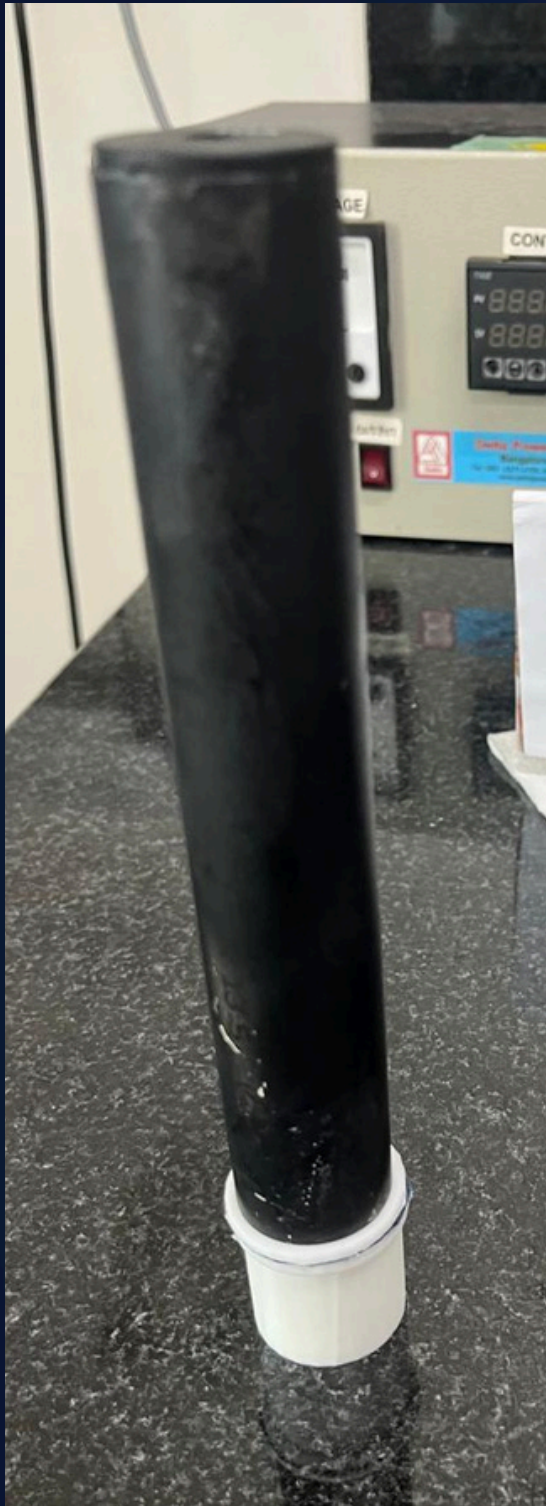


EYEPIECE

a 10x convex lens that slides into the body tube barrel eyepiece diameters are standardized 23 mm for microscopes

BODY TUBE

A 160 mm PVC tube with an inner diameter chosen in such a way that metal exterior of eyepiece passes through inside but plastic cant, painted matte black inside to avoid internal reflections along with glares(bright spots on eyepiece) and enhances the image contrast



SIDE LIGHT



used to illuminate sample from sideways to increase contrast in the image due to body tube shadow alling on sample

OTHER FUN AND LEARNING EXPERIENCES

- *Observed the soldering process for soldering 2 different wires or the ring light*
- *Learning how a lathe machine works - after roughly cutting the pvc tubes in the pipe shop, we went to the workshop and had a hands on experience on the lathing machine sand lathed 4 tubes. (8 sides)*
- *Spray painted matte black paint inside the PVC tubes*
- *Designed and printed CAD models in the solid edge and 3D printers respectively*
- *went to biotech lab to prepare samples that are viewable at 40X magnification*

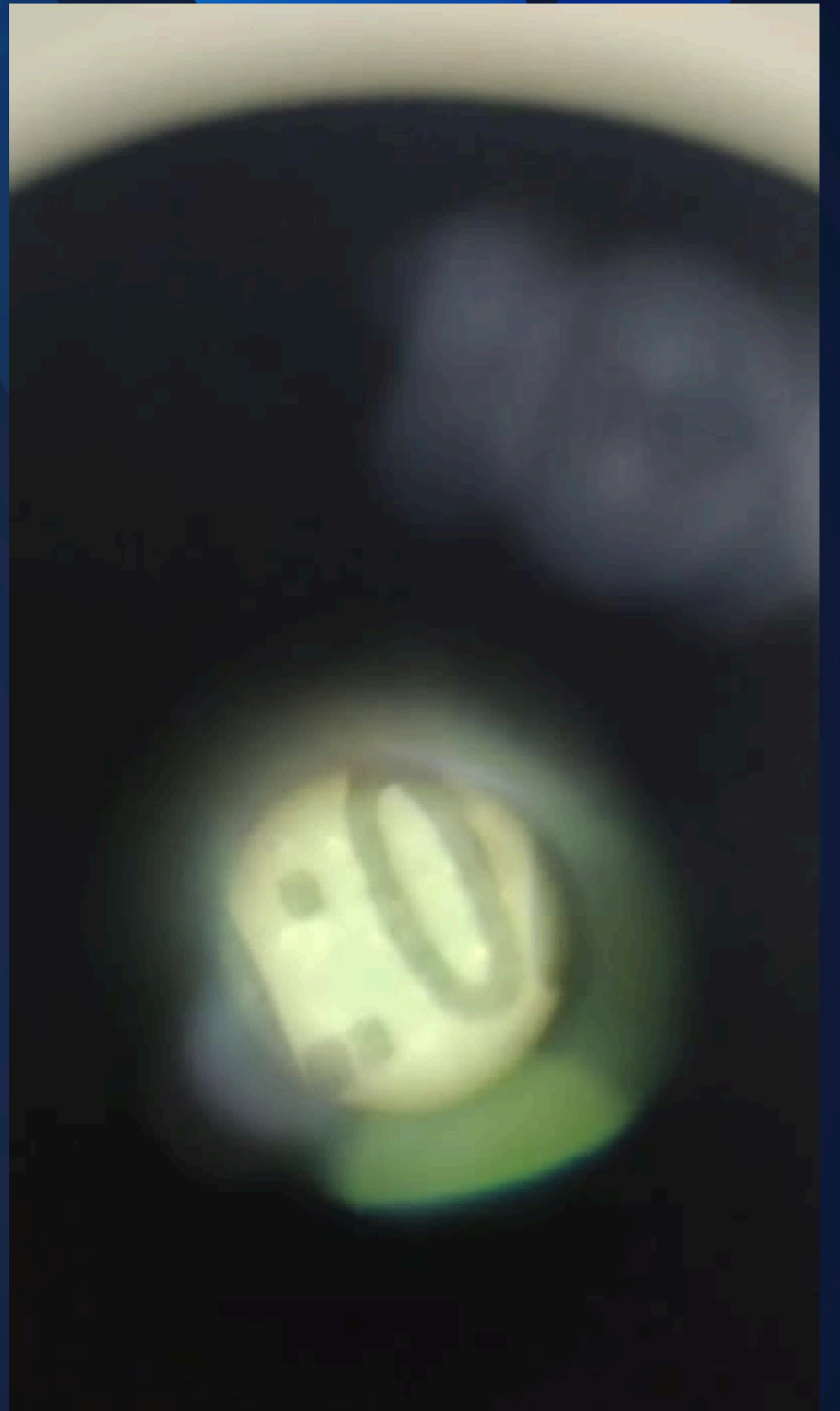
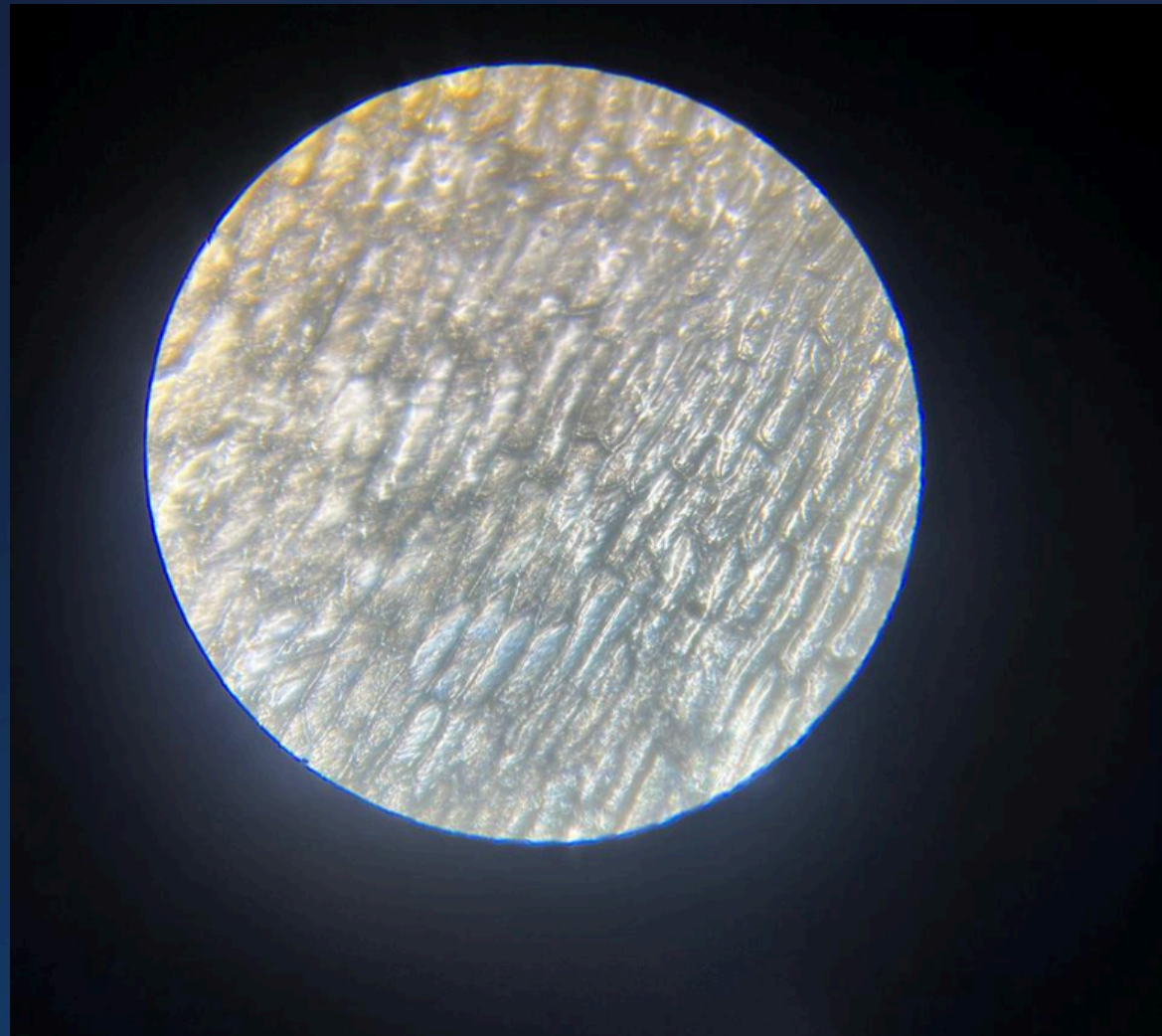
THE FINAL PRODUCT

- The practical magnification achieved was 37.413X. It was calculated using pixels and graph paper and a phone cameras magnifying power.
- The numerical aperture of the objective was 0.1 from there we get the half angle alpha as 5.74 degrees
- The resolution is $0.61 \times \text{wavelength} / \text{Numerical aperture}$ which turned out to be 3.35 microns
- The depth of field was calculated as $\text{wavelength} / \text{numerical aperture squared}$ which came out as 55 micron
- The contrast will be 2-5% for unstained and 25% for stained and around 5-15% for etched specimens
- The focal length for the eyepiece will be $D/M = 250/10 = 25 \text{ mm}$
The focal length for the objective will be $\text{Tube length} / M = 160/4 = 40 \text{ mm}$

THE GRAND ASSEMBLY



AND FINALLY



**THANK YOU ALL FOR YOUR TIME AND FOCUS TODAY—YOUR ATTENTION HAS
MAGNIFIED THE IMPACT OF THIS PRESENTATION.(ONLY PUN INTENDED)**

